Scale-dependent WSPR response to geomagnetic storms

Frequency dependence and exploratory period-to-period scaling of absolute loss

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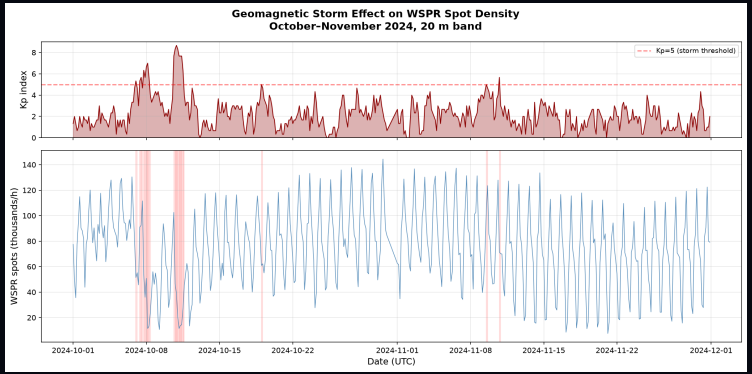
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M. Petrov — co-author, not presenting · Version DOI: 10.5281/zenodo.23402384

Why WSPR?

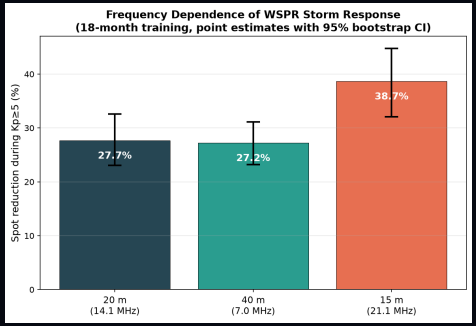
- WSPR (Weak Signal Propagation Reporter): low-rate digital mode used by amateur radio operators worldwide
- Every received spot is an opportunistic probe of one HF propagation path
- Public database: who heard whom, when, on which band
- Global, continuous, free — a distributed ionospheric sensor
- Question: does storm activity suppress HF propagation, as seen in spot counts?

Data and method

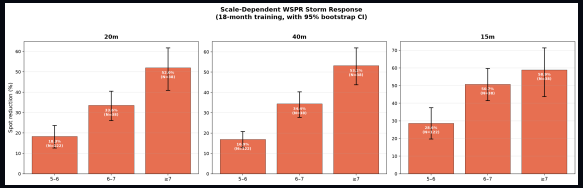


- 18 months, Apr 2024 — Sep 2025; three bands: 20 m, 40 m, 15 m
- 39 456 WSPR hours across the three bands; hourly counts merged 3-hourly with the GFZ Potsdam *Kp* index
- Per band: 4 349 epochs — 198 storm ($Kp \geq 5$) vs 3 035 quiet ($Kp < 3$)
- Block permutation test: 50 000 iterations, block = 24 epochs = 72 h; Benjamini–Hochberg FDR
- Confirmed independently with $Dst < -50$ nT (Kyoto, provisional)

Result 1 — frequency dependence and scale dependence



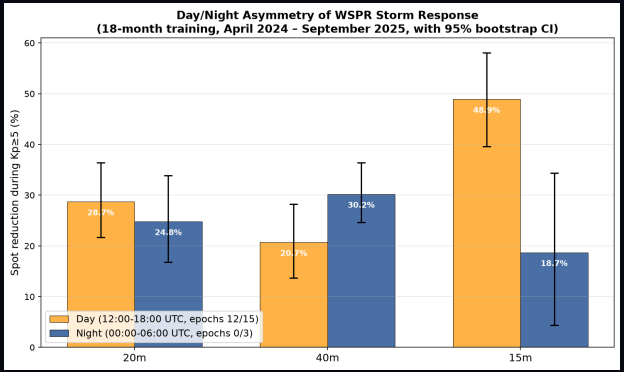
Spot reduction per band



Reduction vs storm intensity (*Kp* bins)

- Raw drops: 20 m $-27.7\,\%$, 40 m $-27.2\,\%$, 15 m $-38.7\,\%$; each $p \leq 2 \times 10^{-5}$
- Covariate-adjusted (month, UT hour, DOW): 20 m $-33.6\,\%$, 40 m $-27.1\,\%$, 15 m $-39.8\,\%$; CIs exclude zero
- Gradient: $\sim -18\,\%$ (*Kp* 5–6) $\rightarrow \sim -34\,\%$ (*Kp* 6–7) $\rightarrow \sim -53\,\%$ (*Kp* ≥ 7); significant trend, highest bin sparse (12 episodes)
- Pattern *consistent with* two mechanisms: MUF reduction (15 m) and D-region absorption (20/40 m)

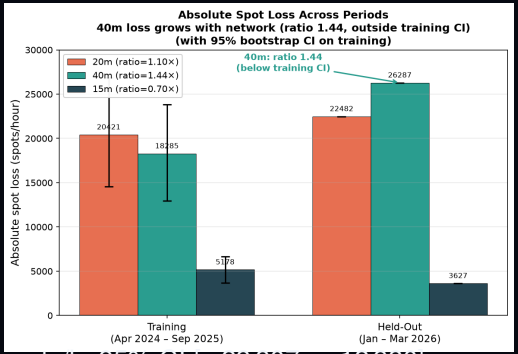
Result 2 — UT-window asymmetry



- UT-day (12–18 UTC): 15 m $-48.9\,\%$ > 20 m $-28.7\,\%$ > 40 m $-20.7\,\%$
- UT-night (0–6 UTC): 40 m $-30.2\,\%$ > 20 m $-24.8\,\%$ > 15 m $-18.7\,\%$
- UT-night excess on 40 m: auroral / polar-cap absorption, scaling as $1/f^2$

Training period. The UT-window asymmetry is based on training data; the held-out window does not independently resolve this (47 storm epochs).

Result 3 — absolute loss on 40 m: invariance?



- Training: $-18\,285$ spots/h, 95% CI $[-23\,807,\ -12\,930]$
- Initial held-out snapshot: $-18\,304$ spots/h $\Rightarrow$ ratio 1.00
- Live re-download (15 Oct 2026): $-26\,287$ spots/h, CI $[-37\,144,\ -12\,472] \Rightarrow$ ratio 1.44 (95% CI [0.65, 2.33])
- Network grew $+35\,\%$ (40 m quiet baseline); expected ratio ≈ 1.35 under proportional scaling
- Ratio CI also includes 1: compatible with proportional scaling, but not conclusively established; percentage drops remain the more stable metric

Held-out replication — Jan–Mar 2026

Band	N_{tot}	N_{storm}	Raw reduction	p-value
20 m	720	47	$-26.7\,\%$ [-41.5, -3.4]	$\leq 2 \times 10^{-5}$
40 m	720	47	$-29.0\,\%$ [-40.8, -14.0]	2×10^{-4}
15 m	720	47	$-22.1\,\%$ [-42.5, 22.8]	8×10^{-5}

- Primary effect replicates in direction on all three bands under the residual permutation test ($p \leq 2 \times 10^{-4}$; 15 m raw CI includes zero)
- Adjusted drops: -19.1, -21.2, $-23.4\,\%$; all CIs exclude zero, but smaller than training — directional, not exact
- Scale-dependence at *Kp* 6–7 agrees within factors 1.01–1.12 (only 10 storm epochs per band: directional check)

Limitations

- 18 months in a solar-maximum phase — not a solar-cycle study
- Held-out power: 47 storm epochs; secondary effects (UT-window asymmetry) not independently re-tested
- 15 m held-out raw 95% CI still includes zero (residual-based p is significant)
- UT windows (12–18 / 0–6 UTC) are not local day/night for a global network
- Distance bins: strongest loss on 5–10 000 km paths; no latitude or local-time resolution yet
- Spot counts conflate propagation with station activity (no active-transmitter normalisation)
- F10.7 only monthly; SIDs not separated; 15 m effect is present in all seasons and not driven by summer sporadic-E
- 198 storm epochs $\approx$ 58 independent episodes (N_{eff}); Dst is provisional
- Decoded-spot SNR changes by < 0.5 dB: the loss is mainly marginal paths dropping below the decoding threshold

Conclusions

1. Storms ($Kp \geq 5$) cut WSPR spot density by 27–39 % raw ($p \leq 2 \times 10^{-5}$); adjusted drops 27–40 %
2. The scale-dependent pattern: ~ -18 % at Kp 5–6, ~ -34 % at Kp 6–7, ~ -53 % at $Kp \geq 7$; trend significant, highest bin sparse
3. It replicates in direction on the held-out Jan–Mar 2026 window (all three bands, $p \leq 2 \times 10^{-4}$)
4. Absolute loss is compatible with network-related scaling (ratio 1.44, 95% CI $[0.65, 2.33]$), but not conclusively established; percentage drops remain the more stable metric
5. Frequency/UT pattern is *consistent with* a dual mechanism: MUF reduction (15 m) and D-region absorption (20/40 m)

WSPR is a viable distributed ionospheric sensor for storm-driven HF degradation.

Questions?

Thank you for listening.

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Concept DOI: 10.5281/zenodo.23285958

Code: github.com/FelixRLEPERS/crosscorr

M. Petrov — co-author, not presenting